BRAC UNIVERSITY

Inspiring Excellence

Assignment_02
Sec_23_16
Summer_2024

Remark: Release date: 5.7.2024

Submission date: 11.07.2024 (6 pm) (online submission because of shortage of time)

Total mark is 45. It will be converted to 20 later

1. Let W be the set of all points from R^3 of the form $(1, x_2, x_3)$. Is this a subspace of R^3? [5]

2. Let $A = \begin{bmatrix} 4 & 1 \\ 3 & 2 \end{bmatrix}$ and consider the following subset V of the 2-dimensional vector space R^2, [10]

   $\{x \epsilon R^2 | Ax = 5x\}$.

   (i) Prove that the subset V is a subspace of R^2

   (ii) Find a basis for V and determine the dimension of V

3. Determine the rank of the following matrix A: [15]

$$A = \begin{pmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{pmatrix}$$

4. Find a basis for the null space of $A = \begin{pmatrix} 1 & 4 & 5 & 2 \\ 2 & 1 & 3 & 0 \\ -1 & 3 & 2 & 2 \end{pmatrix}$ [10]

5. **Vertor Space:** A set $\boldsymbol{V}$ equipped with two binary operations addition and scalar multiplication is called a vector space over the field F, if V satisfies the following 10 axioms, [5]

(i) $u+v \in V \; for \; all \; u, v \in V$

(ii) $u+v=v+u$

(iii) $u+(v+w)=(u+v)+w$

(iv) There exists a $0 \in V \; s.t. \; u+0=u \; for \; all \; u \in V$

(v) There exists a $-u \in V \; for \; all \; u \in V, such \; that \; u+(-u)=0$.

(vi) $ku \in V \; for \; all \; u \in V \; and \; k \in F$

(vii) $a(u+v)=au+av \; for \; all \; a \in F \; and \; u, v \in V$

(viii) $(a+b)u=au+bu \; for \; all \; a, b \in F \; and \; u \in V$.

(ix) $(ab)u=a(bu) for \; all \; a, b \in F \; and \; u \in V$

(x) $1u=u, where \; 1 \in F \; and \; for \; all \; u \in V$.

(a) Suppose $u=(u_1, u_2)$ the multiplication of cu is defined to produce $(cu_{1,}0)$ instead of (cu_1, cu_2). With usual addition in $\mathbb{R}^2$, which of the eight conditions are not satisfied.?